41937 Structural design of precast concrete buildings

Lecture 3

Vertical Members – columns and walls

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DTU Construct

Content

- Columns (and walls) in ULS (and SLS)

General course objectives

The students will be able to carry out a structural design and analysis of a multi-storey building consisting of precast concrete elements.

Learning objectives

A student who has met the objectives of the course will be able to:

- Explain the structural system of a multi-storey precast concrete building.
- Identify and calculate relevant loads for a multi-storey building.
- Determine the load path for vertical load to the foundation.
- Choose and validate structural members in the serviceability and ultimate limit state for vertical loads.
- Determine the load path for horizontal load to the foundation and determine the stabilising system.
- Design and validate the floor diaphragms for distribution of horizontal loads.
- Evaluate the stability of a multi-storey precast concrete building subjected to vertical and horizontal loads.
- Analyze and design the structural joints between the precast elements.
- Validate the load carrying capacity of a wall element by use of relevant numerical tools.

Outline

- From last time
- Background for buckling analysis of vertical concrete members (we have reserved the word ‘stability’ for other phenomena)
- Practical design tool: design diagrams
- Example

Basis for today's lecture

Compendium

DTU Course 11320: Structural analysis of buildings

Autumn 2010

Translations from "Betonelementbyggeriers statik"

Polyteknisk Forlag, 2010

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Euler-column – not relevant due to the presence of eccentricities and the non-linear behavior of concrete columns :

$$u_z(x) = a\,\sin(kx)$$

$$k^2 = \frac{P}{EI_y}$$

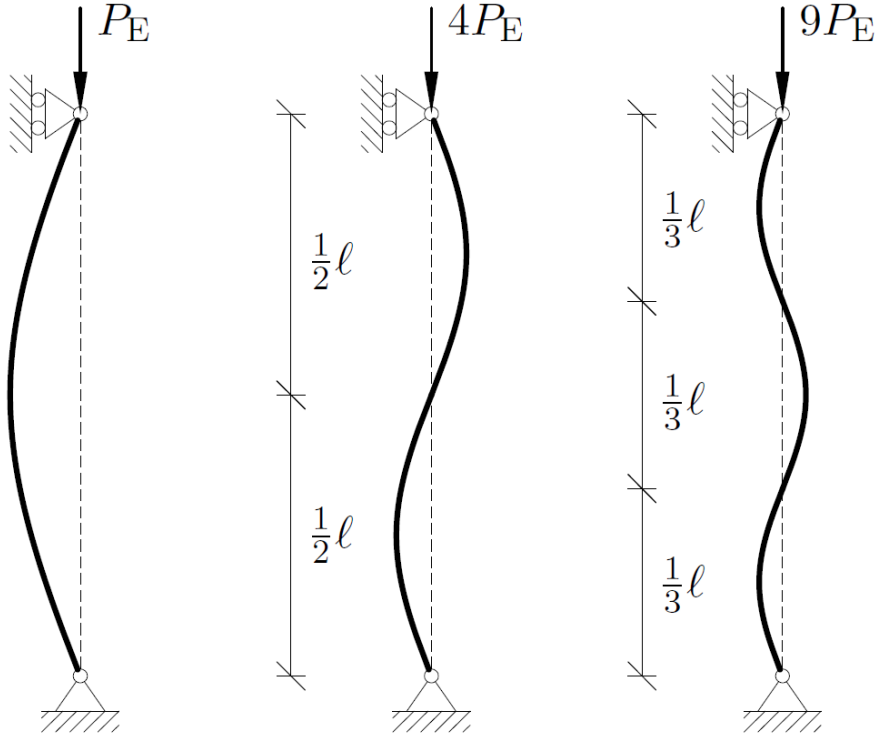
$$\sin(k\ell) = 0 \quad \Rightarrow \quad k\ell = n\pi, \quad n = 1, 2, 3, \ldots$$

$$P_n = k^2 EI_y = n^2 \frac{\pi^2 EI_y}{\ell^2}$$

$$u_z^{(n)}(x) = a_n\,\sin\left(n\pi\frac{x}{\ell}\right)$$

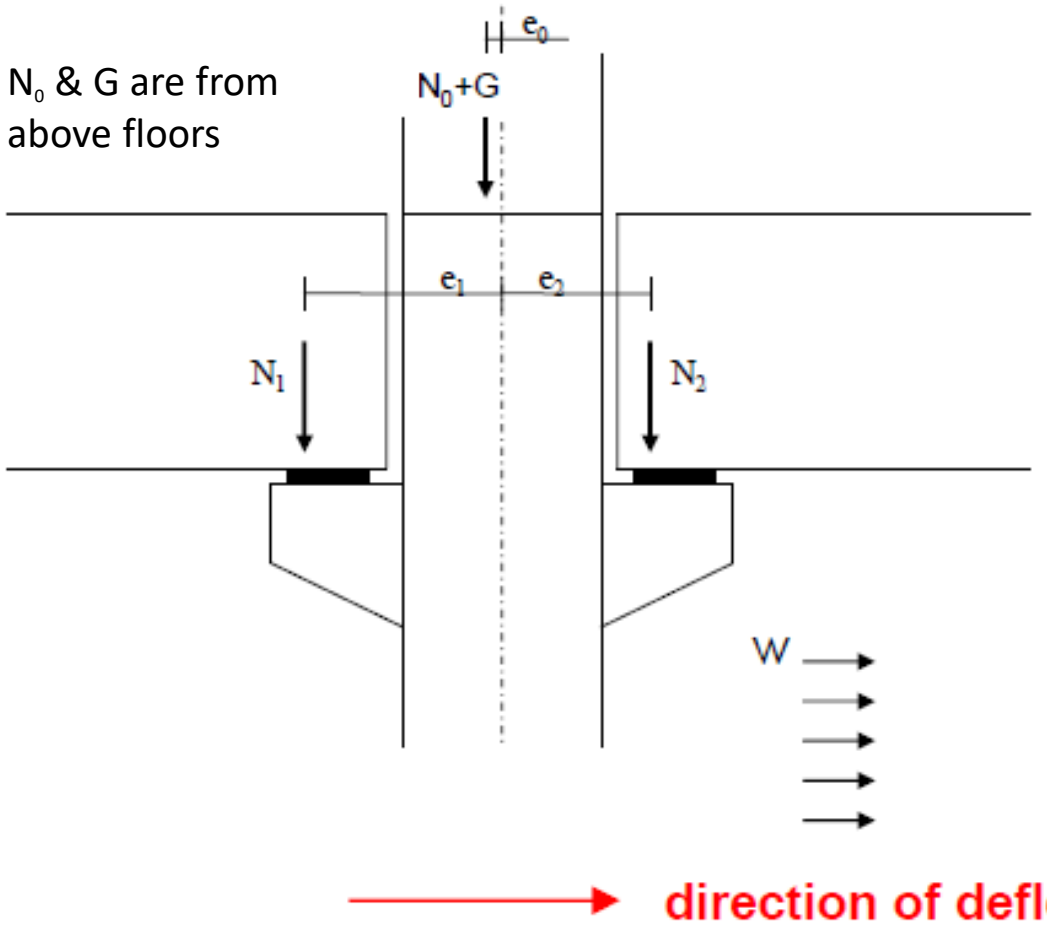
for *n* = 1 fås Euler-lasten:

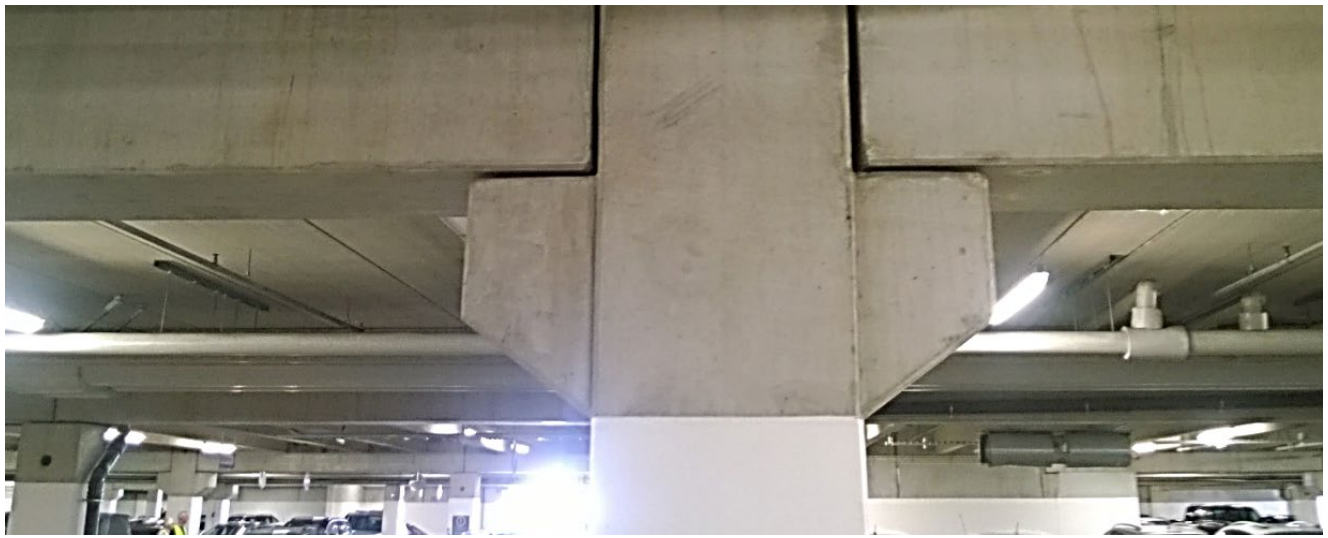
$$P_{\mathrm{E}} = \frac{\pi^2 EI_y}{\ell^2}$$

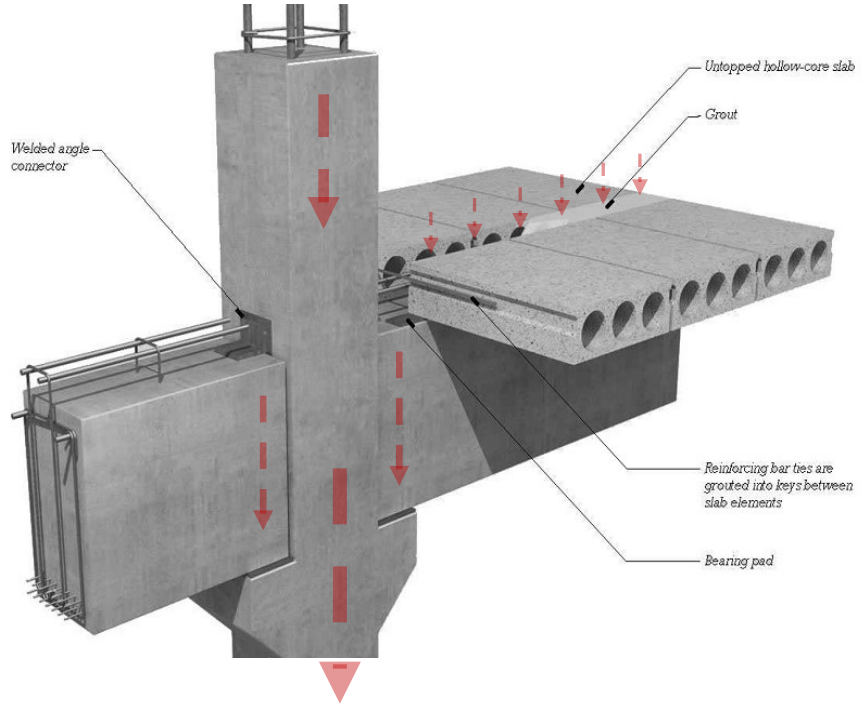


Eccentricities

Eccentricities in columns

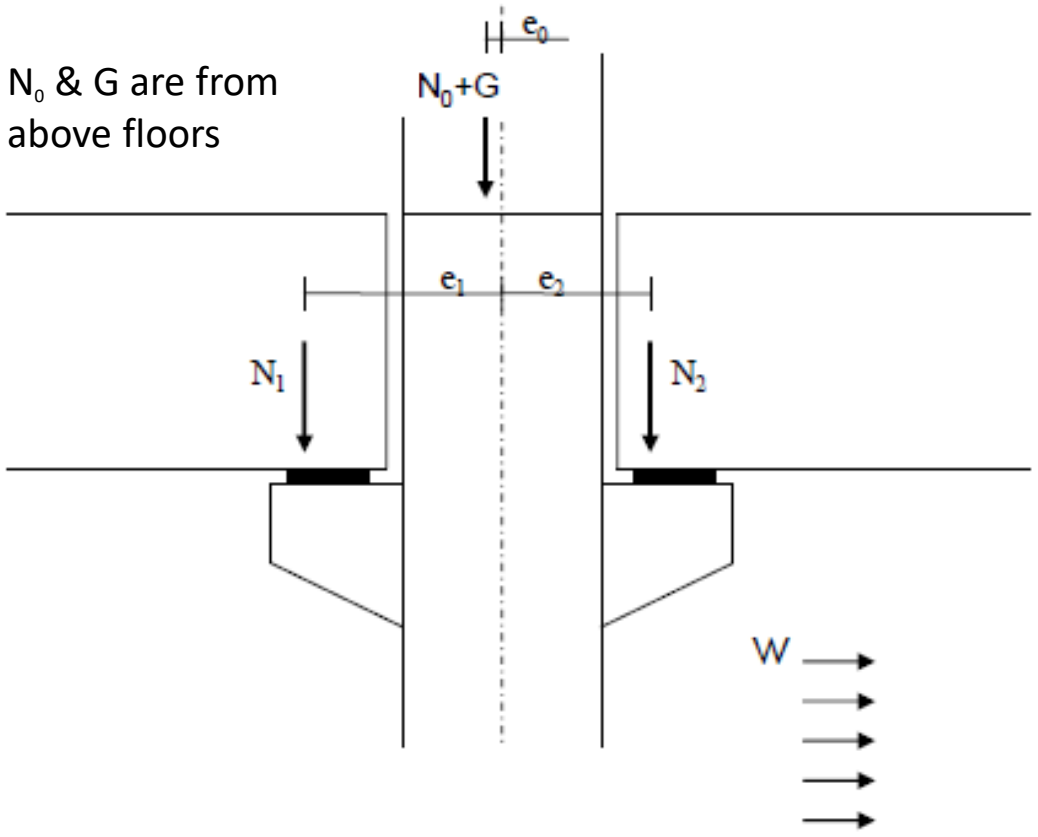


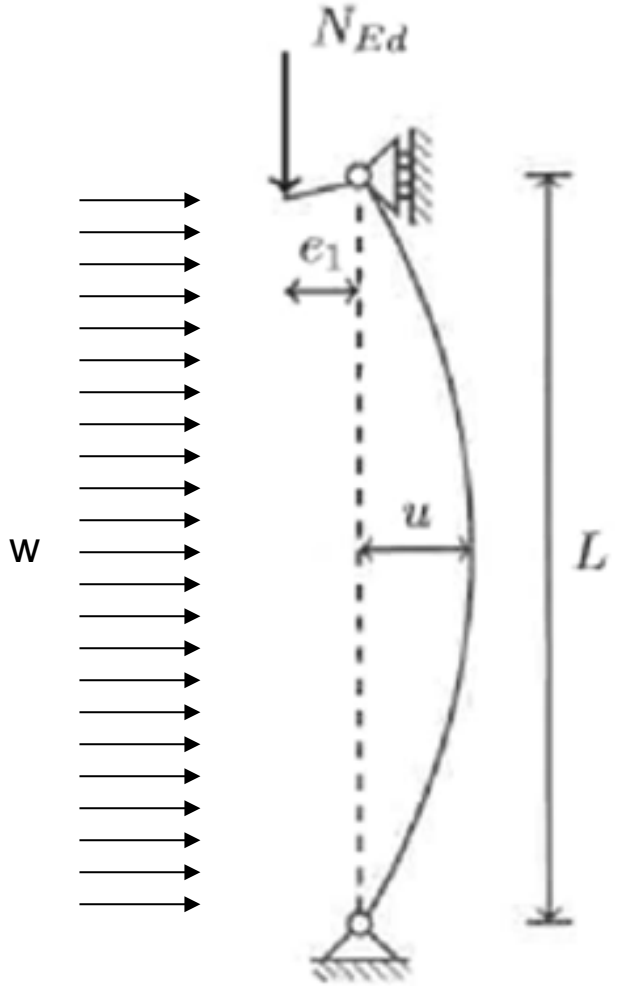




Eccentricities

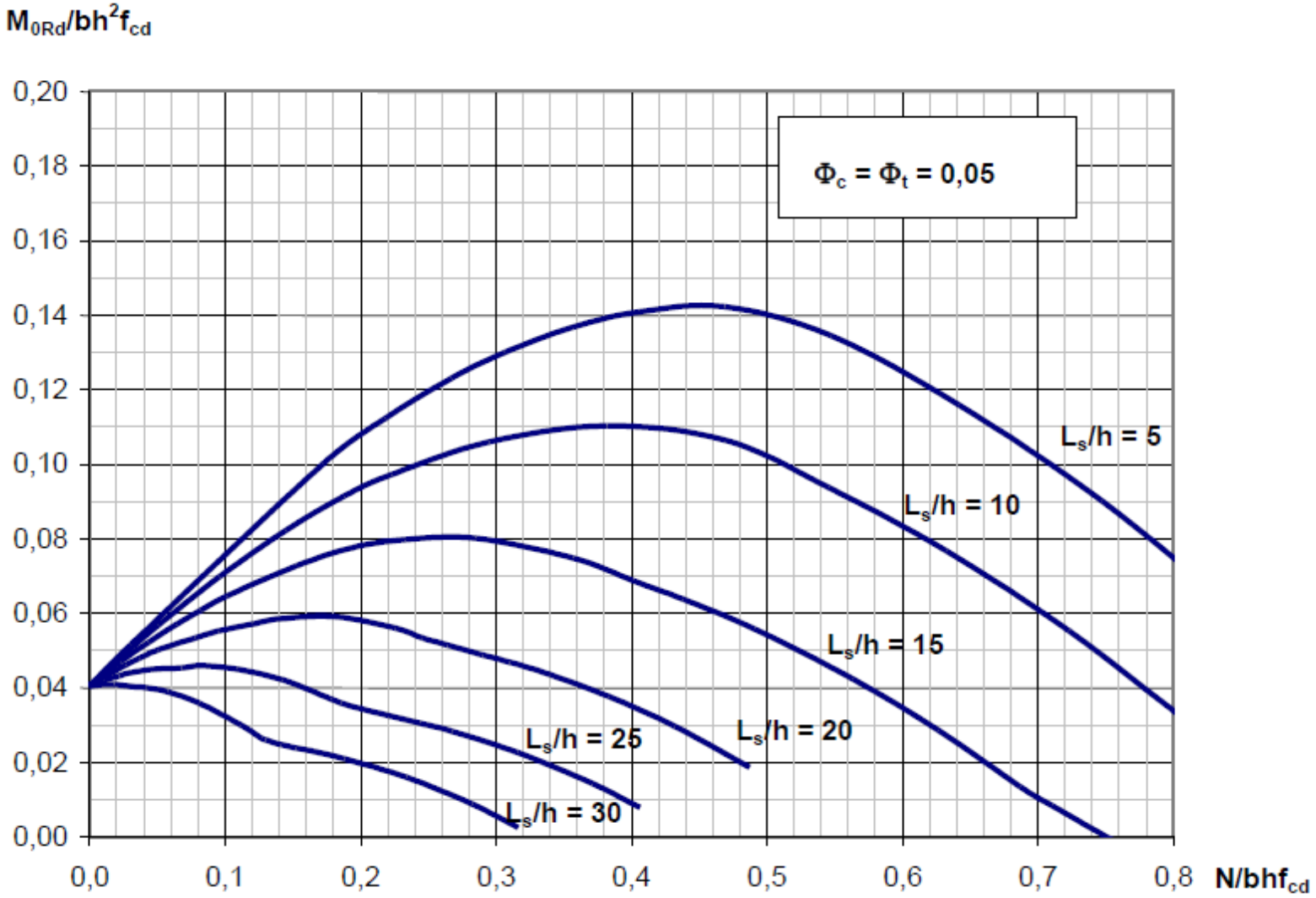
Eccentricities in columns



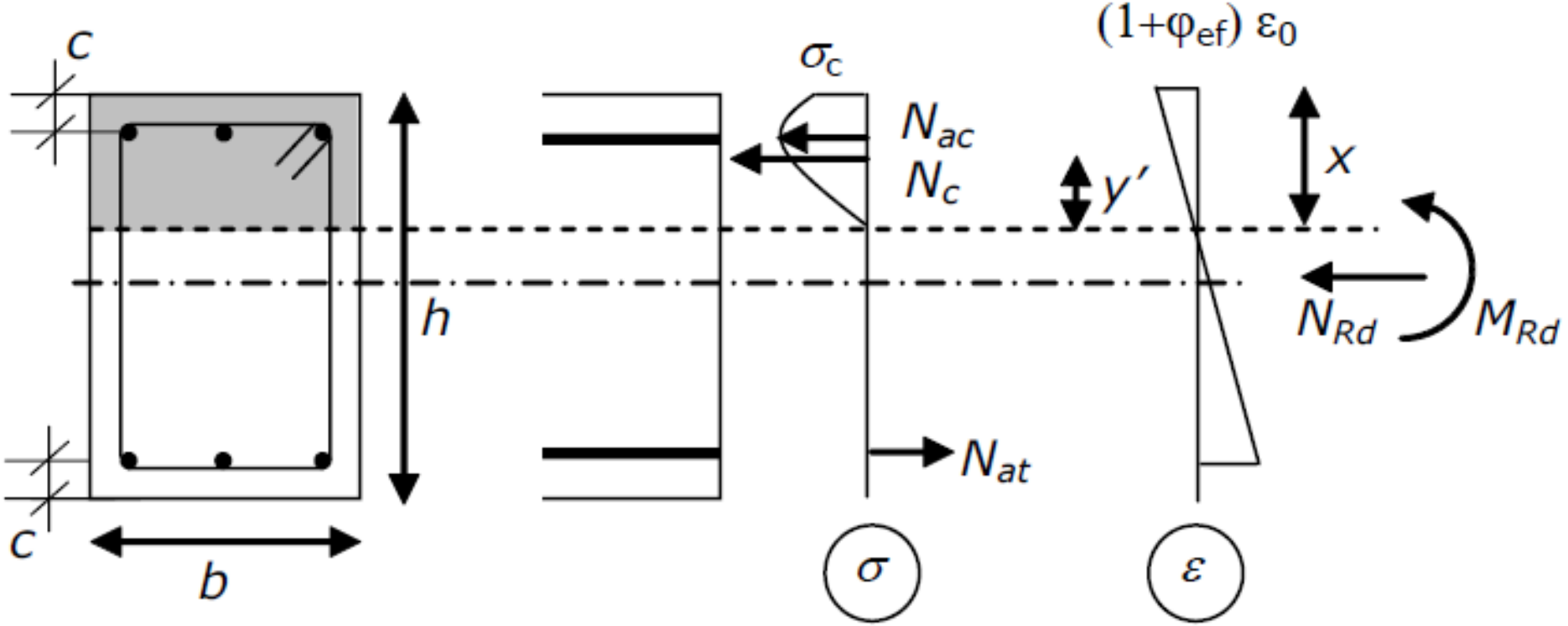


direction of deflection

Combined normal force and moment



Combined normal force and moment



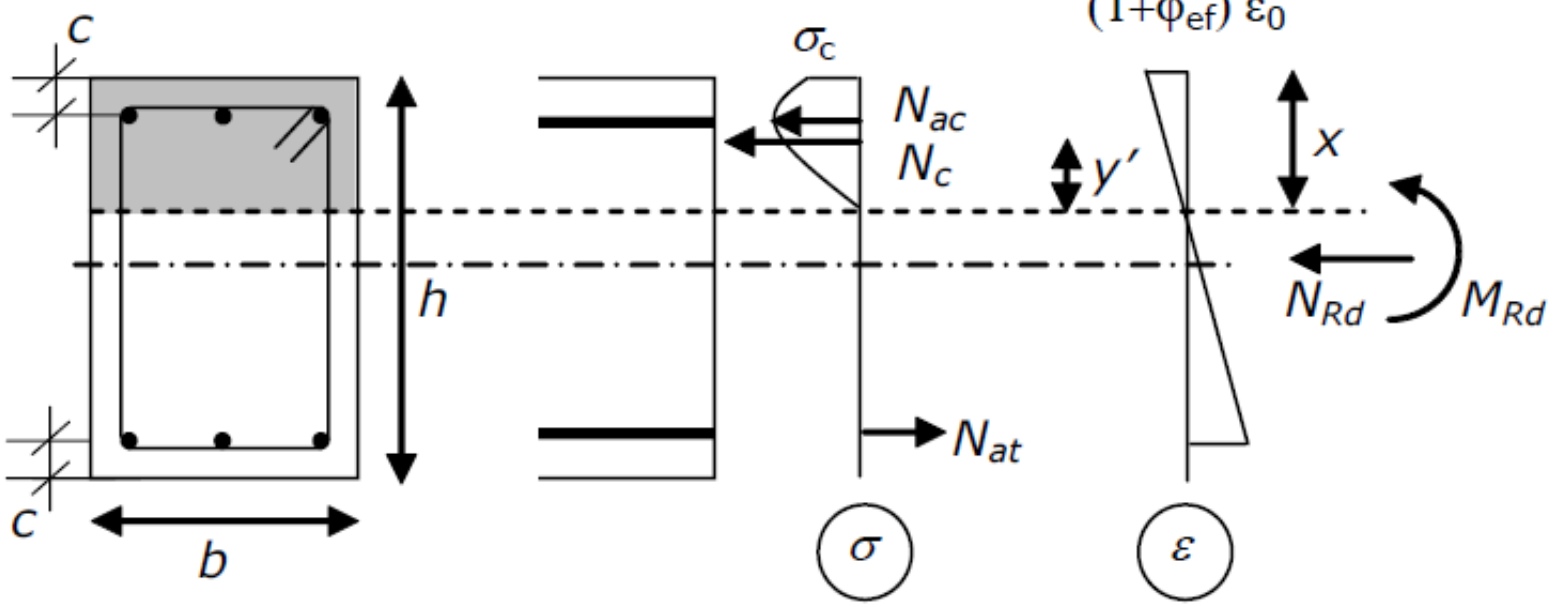
Creep

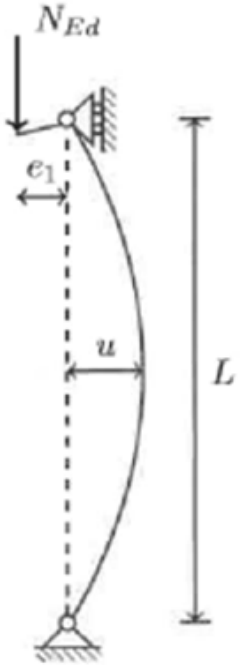
$$\varepsilon_{cc,\infty} = (1+\varphi_0)\cdot\varepsilon_c$$

$$\varphi_0 = \varphi_0(t_0, RH, f_c, h_0, ct)$$

$$\varphi_{ef} = \frac{\text{moment from long term load}}{\text{moment from total load}}\varphi_0$$

Combined normal force and moment





1st order moment (from excentric normal forces)

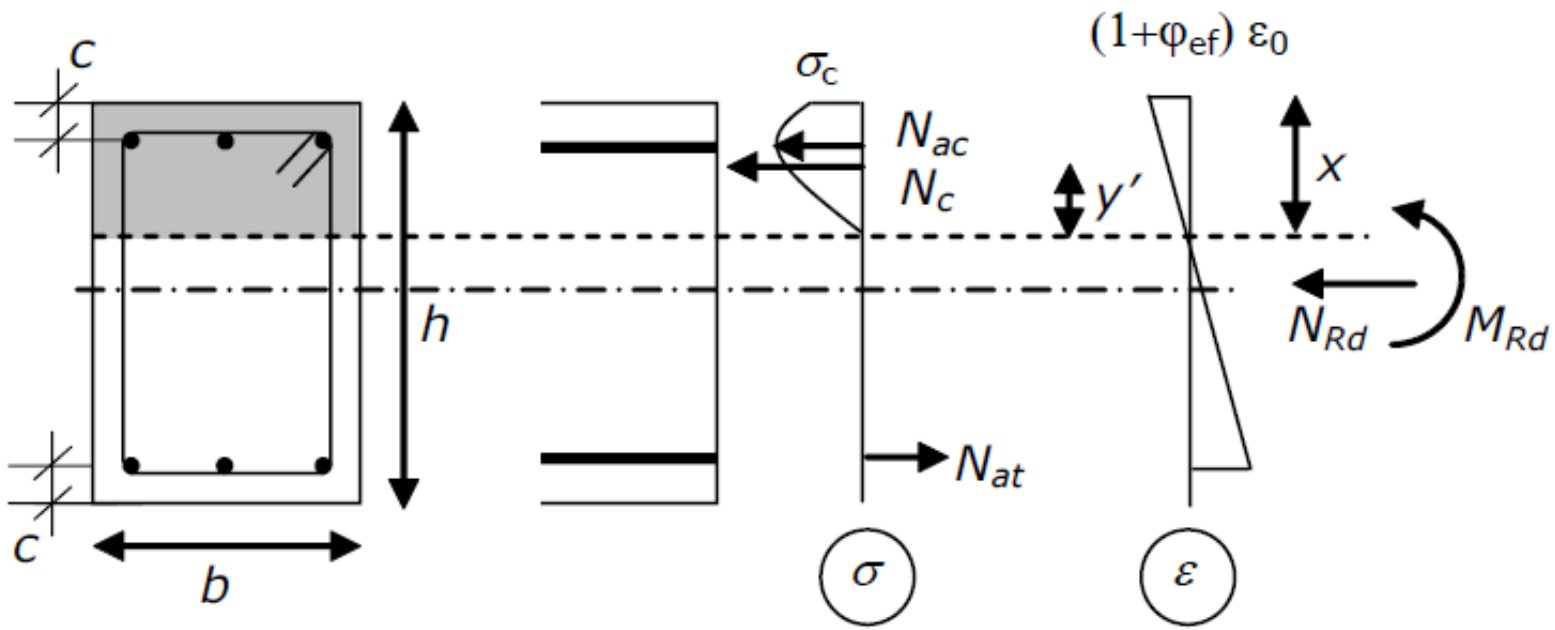
2nd order moment (from deflection)

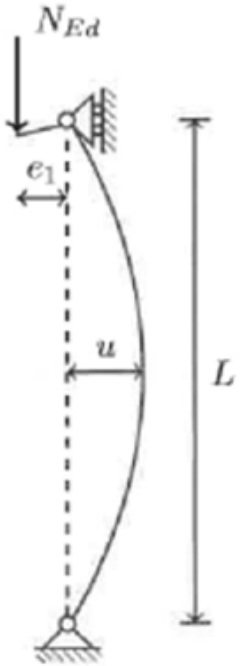
$$M_{Rd} = M_{0Rd} + N_{Ed}u$$

$$N_{Rd} = N_{Ed} = \sum_i N_{id}$$

$$M_{0Rd} = \sum_i N_{id} e_i$$

Combined normal force and moment





1st order moment 2nd order moment

$$M_{Rd} = M_{0Rd} + N_{Ed}u \Leftrightarrow$$

$$M_{0Rd} = M_{Rd} - N_{Ed}\frac{1}{10}\frac{\left(1+\varphi_{ef}\right)\varepsilon_0}{x}L_s^2$$

$$u \cong \frac{1}{10}\kappa_{\max}L_s^{\ 2}$$

$$M_{0Rd} = \sum_i N_{id}e_i$$

Combined normal force and moment – determine capacity

For a given ε_0 determine *x* from: $N_{Ed} = N_c + N_{ac} - N_{at}$

The compressive force in the concrete N_c is determined for a given ε_0 as a function of x using simplified representation of the stress distribution in ULS, taking creep into account

The compressive reinforcement: $$N_{ac} = \min\begin{cases} \frac{x-c}{x}\left(1+\varphi_{ef}\right)\varepsilon_0 A_{sc} E_s \\ A_{sc} f_{yd} \end{cases}$$

The tensile reinforcement: $$N_{at} = \min\begin{cases} \frac{h-x-c}{x}\left(1+\varphi_{ef}\right)\varepsilon_0 A_{st} E_s \\ A_{st} f_{yd} \end{cases}$$

Combined normal force and moment – determine capacity

For a given ε_0 determine *x* from: $N_{Ed} = N_c + N_{ac} - N_{at}$

The total moment capacity can be determined from: $M_{Rd} = \left(\frac{1}{2}h - x + y'\right)N_c + \left(\frac{1}{2}h - c\right)N_{ac} + \left(\frac{1}{2}h - c\right)N_{at}$

Combined normal force and moment – determine capacity

For a given ε_0 determine *x* from: $N_{Ed} = N_c + N_{ac} - N_{at}$

The total moment capacity can be determined from: $M_{Rd} = \left(\frac{1}{2}h - x + y'\right)N_c + \left(\frac{1}{2}h - c\right)N_{ac} + \left(\frac{1}{2}h - c\right)N_{at}$

The 1st order moment capacity can be determined from: $M_{0Rd} = M_{Rd} - N_{Ed}\frac{1}{10}\frac{\left(1+\varphi_{ef}\right)\varepsilon_0}{x}L_s^2$

ε_0

Combined normal force and moment – determine capacity

For a given ε_0 determine *x* from: $N_{Ed} = N_c + N_{ac} - N_{at}$

The total moment capacity can be determined from: $M_{Rd} = \left(\frac{1}{2}h - x + y'\right)N_c + \left(\frac{1}{2}h - c\right)N_{ac} + \left(\frac{1}{2}h - c\right)N_{at}$

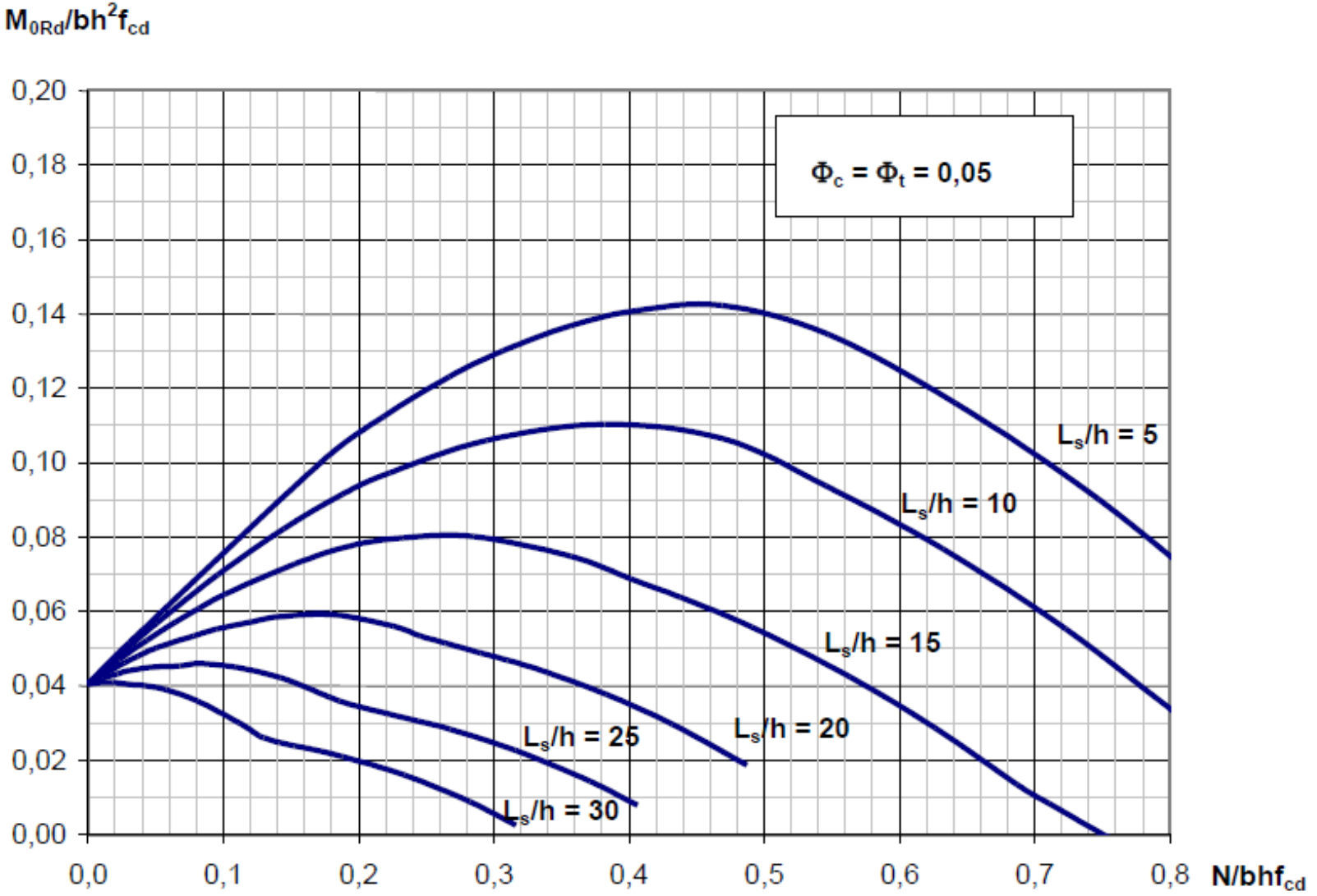
The 1st order moment capacity can be determined from: $M_{0Rd} = M_{Rd} - N_{Ed}\frac{1}{10}\frac{\left(1+\varphi_{ef}\right)\varepsilon_0}{x}L_s^2$

Different values of ε_0 are investigated to find the largest 1st order moment capacity

Combined normal force and moment – design diagrams

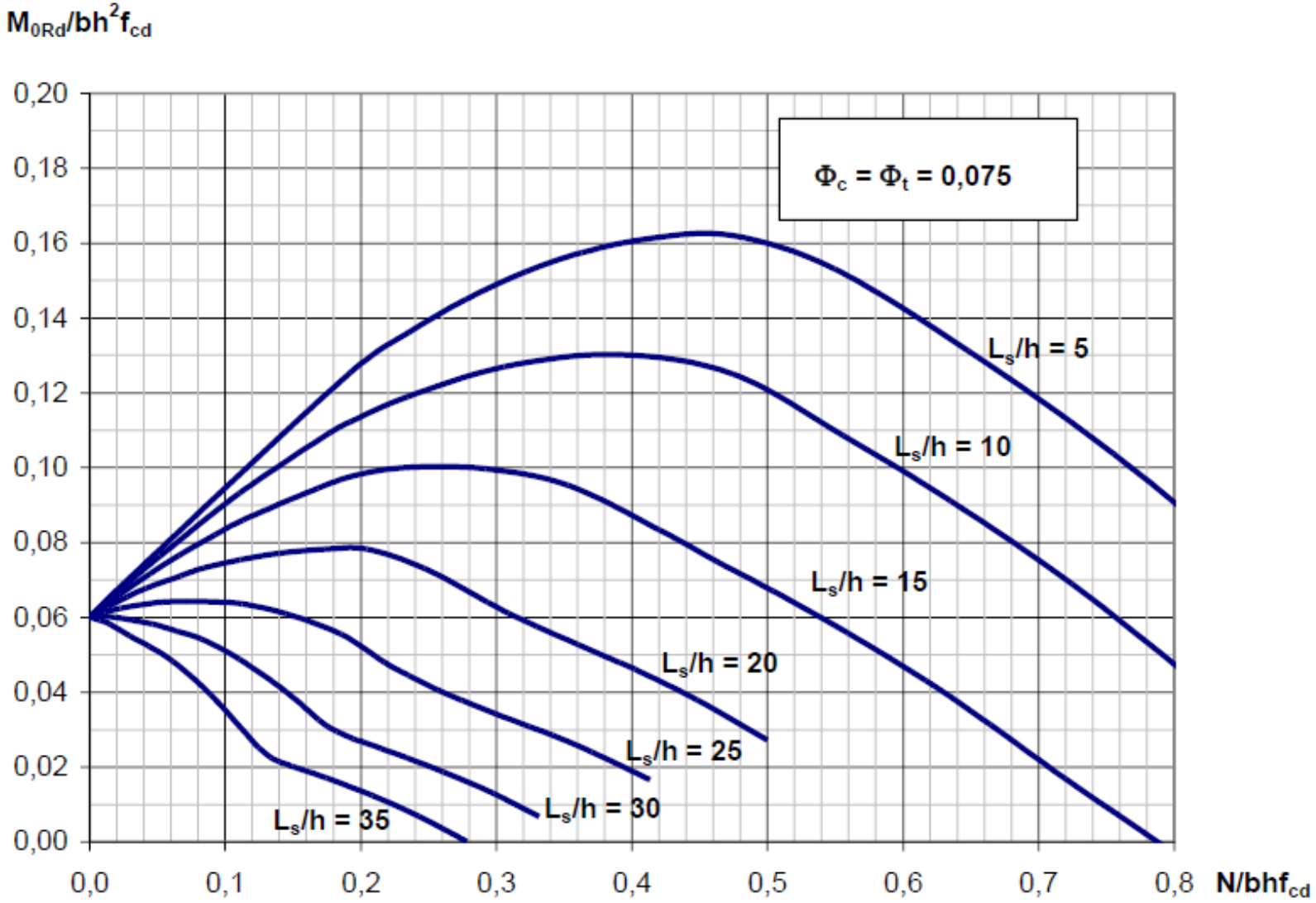
Cross section form:	Apply to rectangular cross sections
Concrete strength:	$20\ \text{MPa} \leq f_{ck} \leq 50\ \text{MPa}$
Reinforcement strength:	$400\ \text{MPa} \leq f_{ck} \leq 600\ \text{MPa}$
Distance from concrete edge to centre of main reinforcement, $c \leq h/10$	
Effective creep coefficient:	φ_{ef} = 1,6 corresponding to dry indoor climate. Lower values of the effective creep coefficient give bearing capacities on the safe side.

Combined normal force and moment – design diagrams for columns for various reinforcement ratios



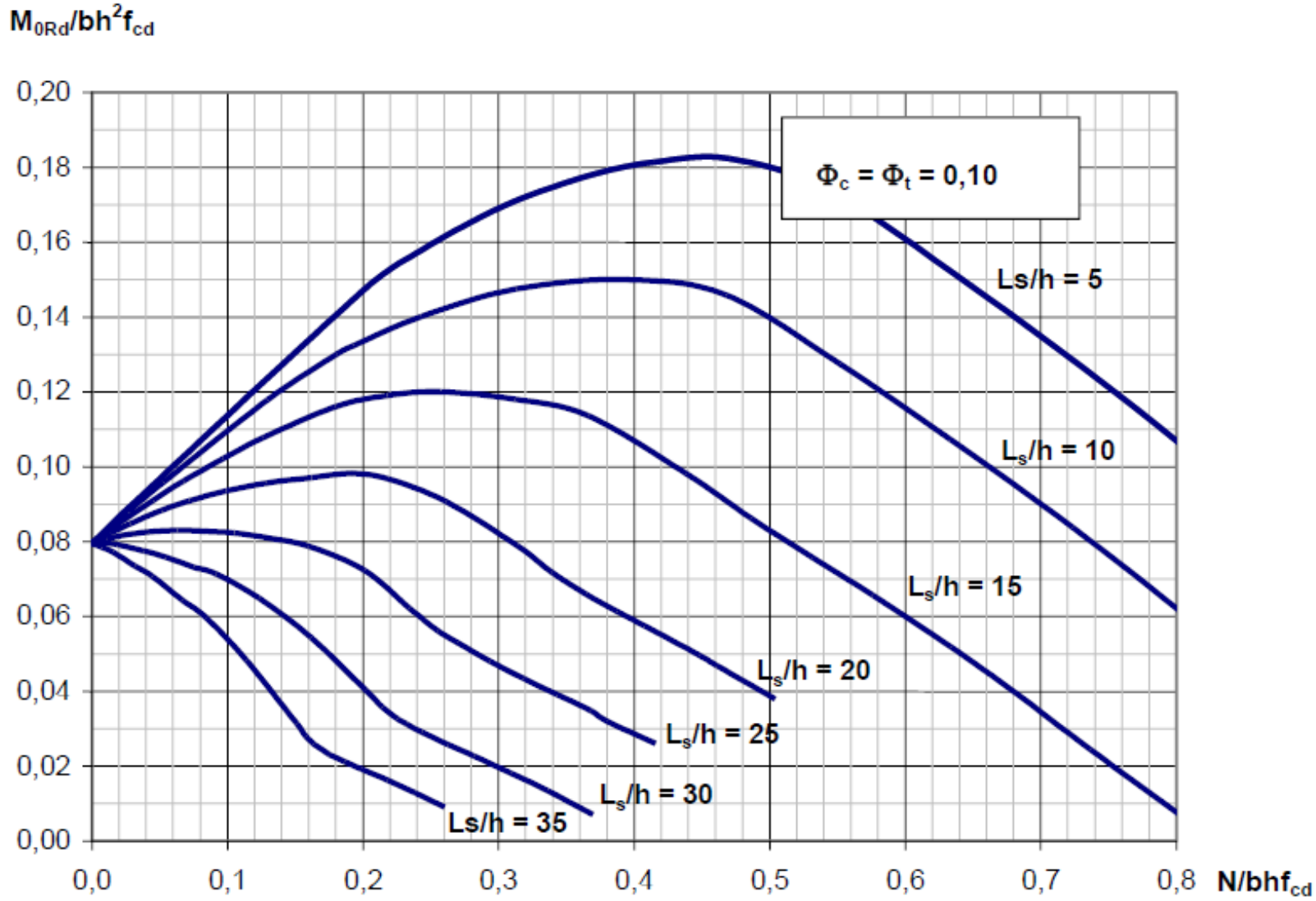
$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

Combined normal force and moment – design diagrams for columns for various reinforcement ratios



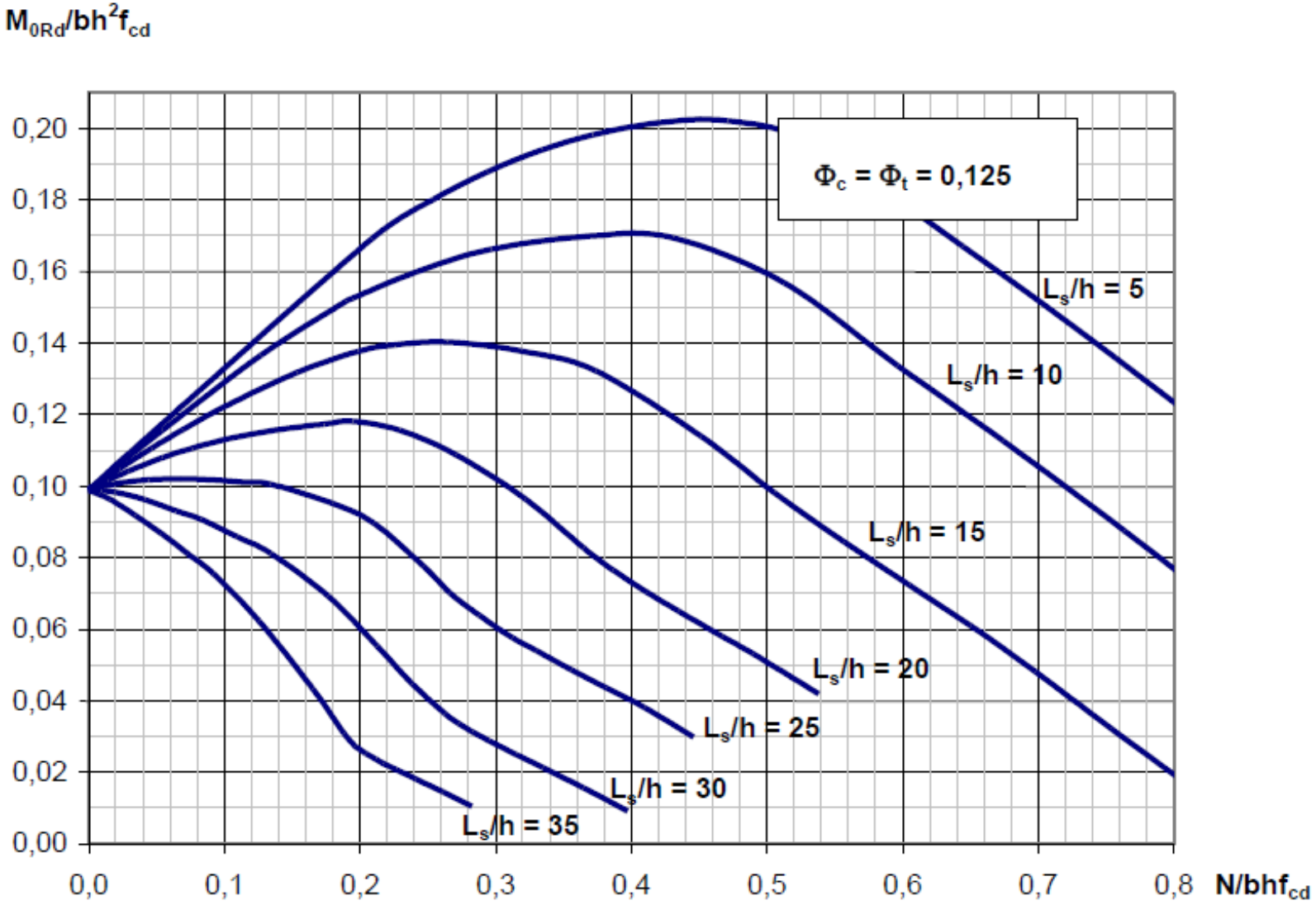
$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

Combined normal force and moment – design diagrams for columns for various reinforcement ratios



$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

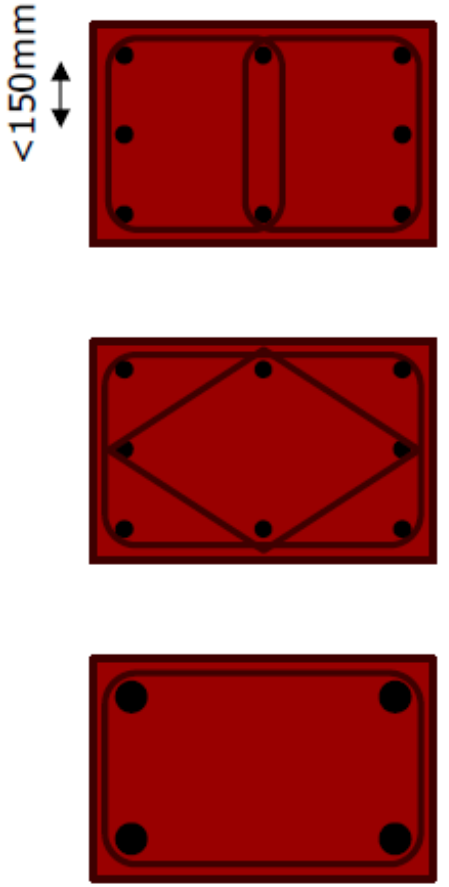
Combined normal force and moment – design diagrams for columns for various reinforcement ratios



$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

Rules of thumb in design of vertical members

- Never a smaller dimension than Ø6 in walls and Ø8 in columns.
- At least one bar in each corner, and no more than 400mm between bars.
- At least 4 bars in a round column.
- Rods in corners of columns must be enclosed by braces (for retention). Vertical bars that are not at a corner must not be more than 150mm from a hanger.



$$A_{s,\min} = \max\left\{\begin{array}{l}\dfrac{0,10N_{Ed}}{f_{yd}}\\ 0,002A_c\end{array}\right\} \qquad A_{s,\max} = 0,04A_c$$

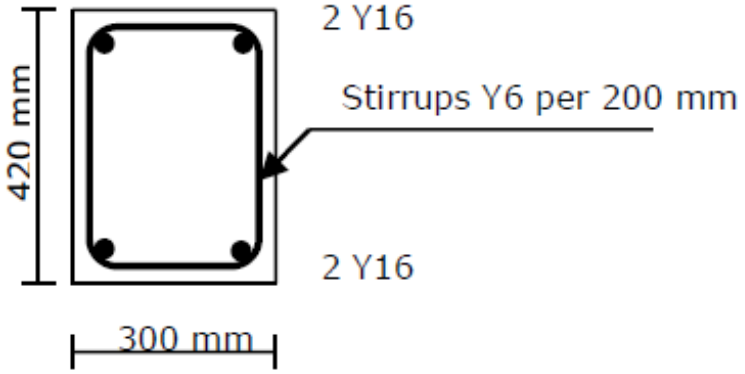
Combined normal force and moment – practical use of design diagrams for verification

Calculation premises

Cross section 420 mm x 300 mm

Characteristic concrete compressive strength	f_{ck} = 35 MPa
Design value of concrete compressive strength	f_{cd} = 35 MPa/1,4 = 25MPa
Reinforcement 4 pieces Y16, one in each corner:	A_{sc} = 402 mm^2
	A_{st} = 402 mm^2
	c = 40 mm
Characteristic yield strength	f_{yk} = 500 MPa
Design value of yield strength	f_{yd} = 500 MPa/1,2 = 417 MPa

Column length L_s = 3500 mm around both axes.

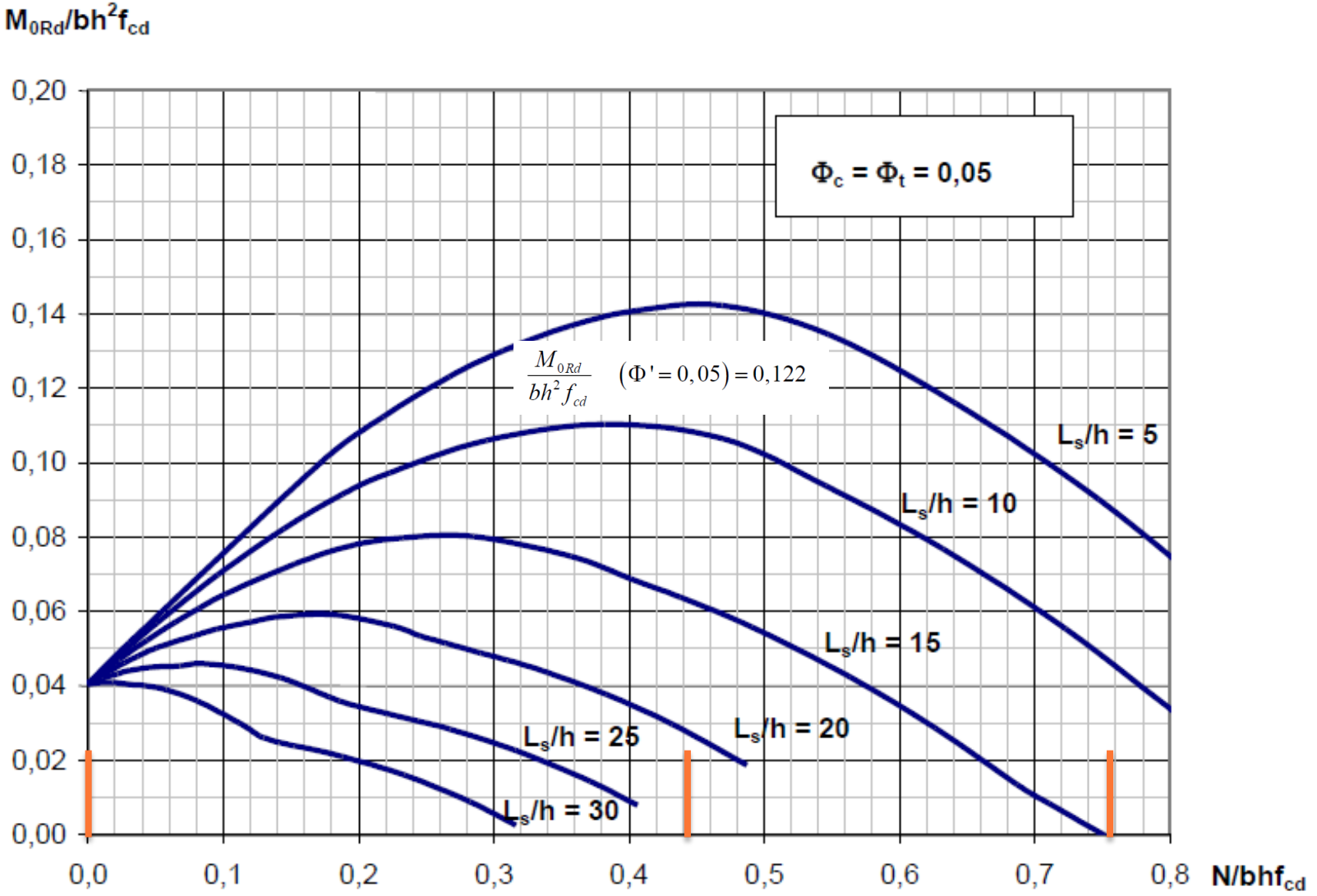


$$\Phi_t' = \Phi_c' = \frac{Af_{yd}}{bhf_{cd}} = \frac{402 \cdot 417}{300 \cdot 420 \cdot 25} = 0,053$$

$$\frac{L_s}{h} = \frac{3500}{420} = 8,3$$

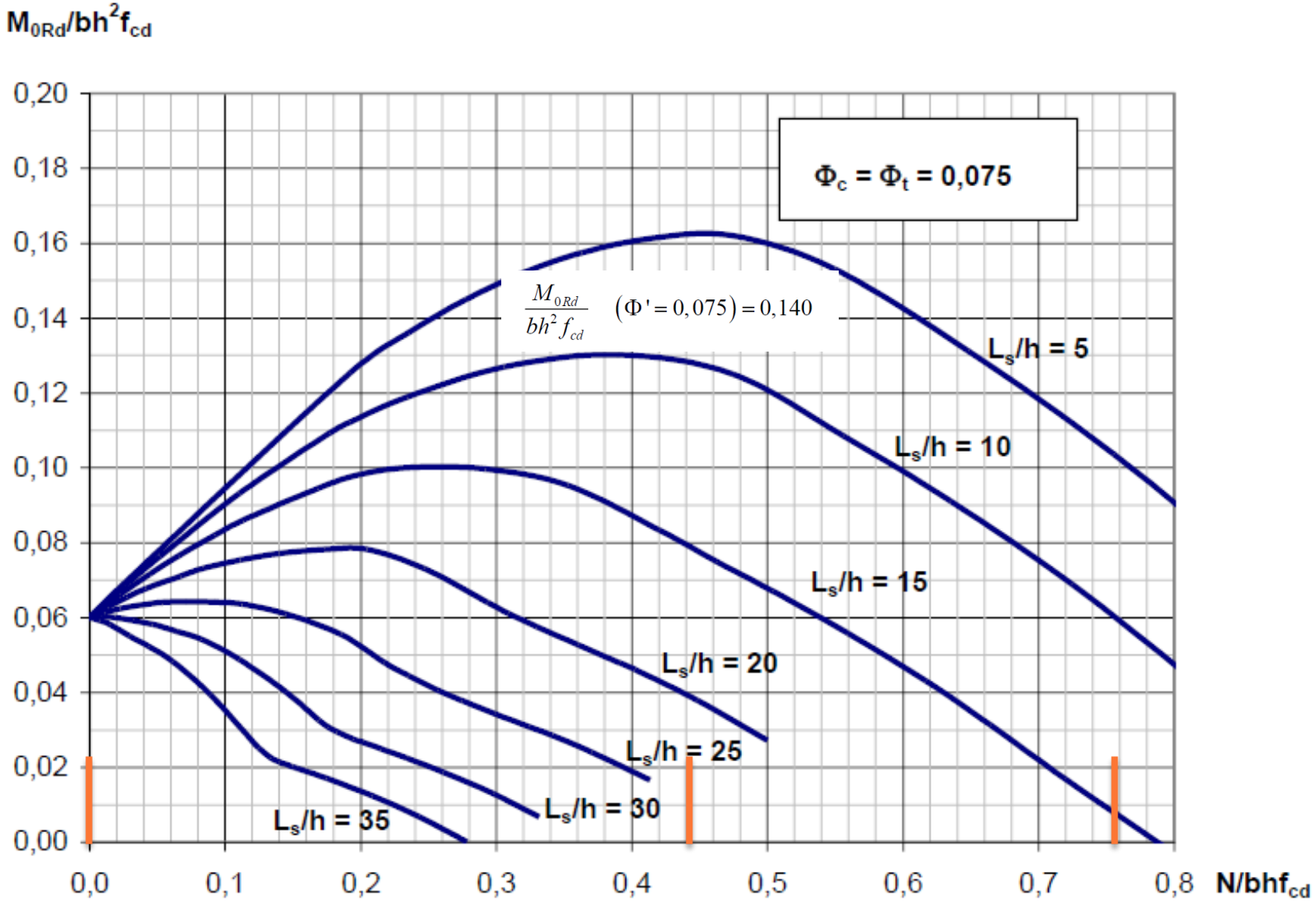
(Strong axis)

Combined normal force and moment – design diagrams for columns for various reinforcement ratios



$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

Combined normal force and moment – design diagrams for columns for various reinforcement ratios



$$\Phi_t' = \frac{A_{st} f_{yd}}{bhf_{cd}}$$

Combined normal force and moment – practical use of design diagrams for verification

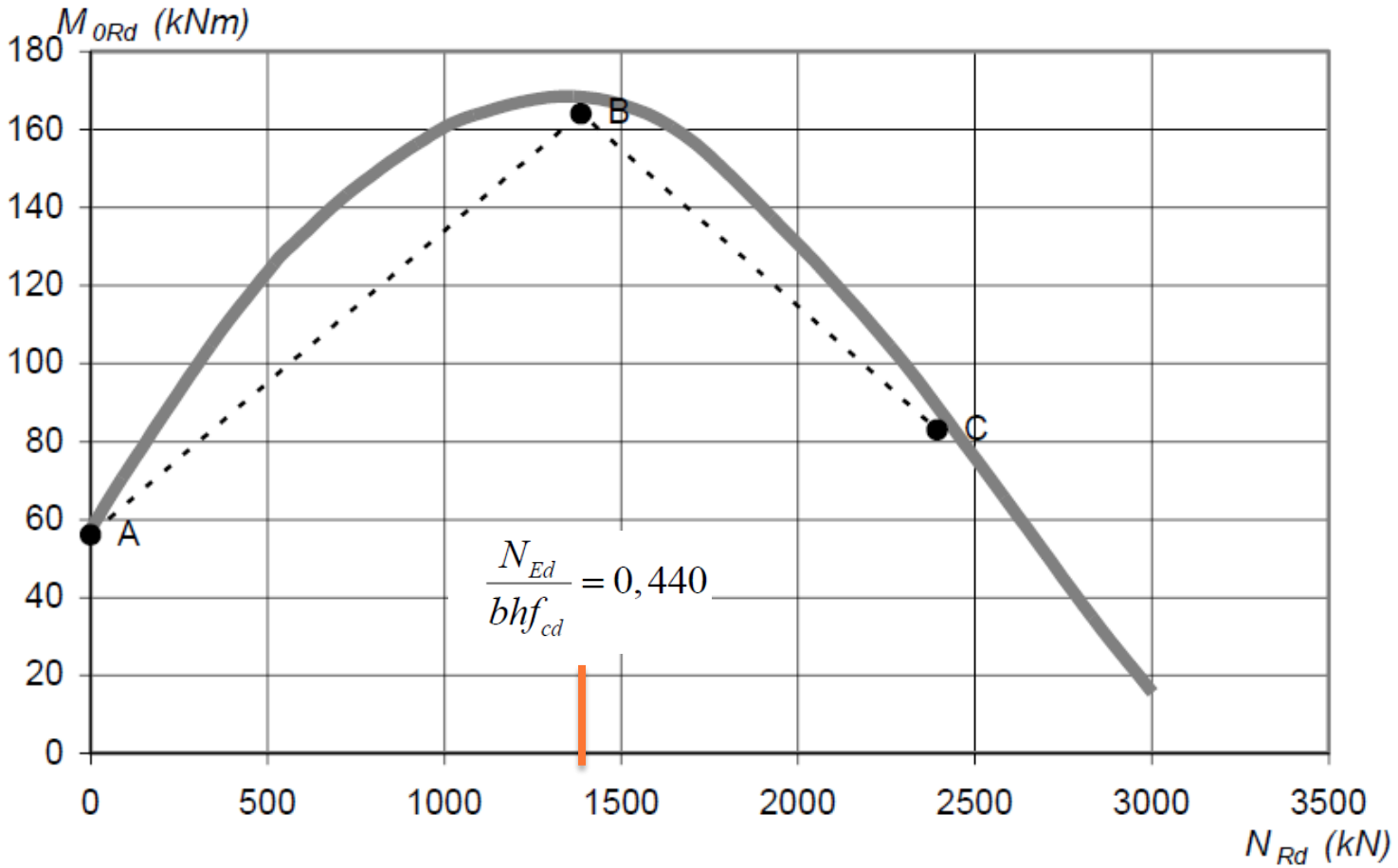
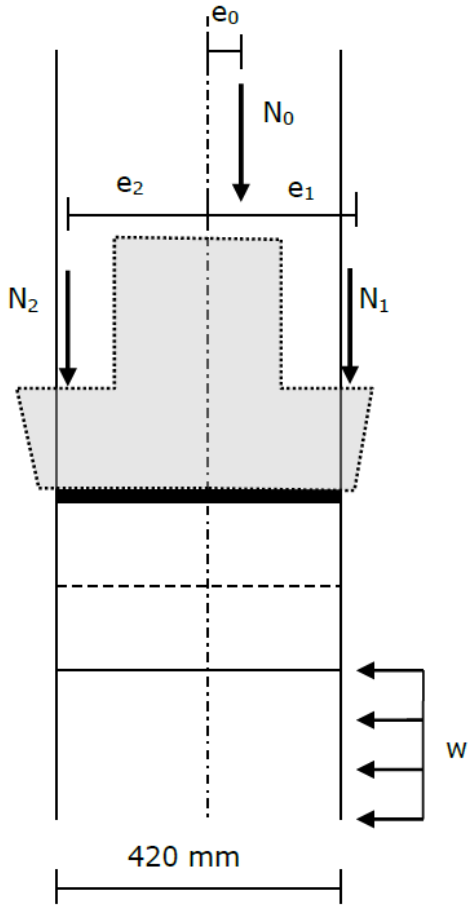


Figure 8-13: Bearing capacity curve (www.bef.dk, Søjleelementer) and approximated bearing capacity curve, bending around strong axis (dotted)

Combined normal force and moment – practical use of design diagrams for verification



$$N_{Ed} = N_1 + N_0 + N_2$$

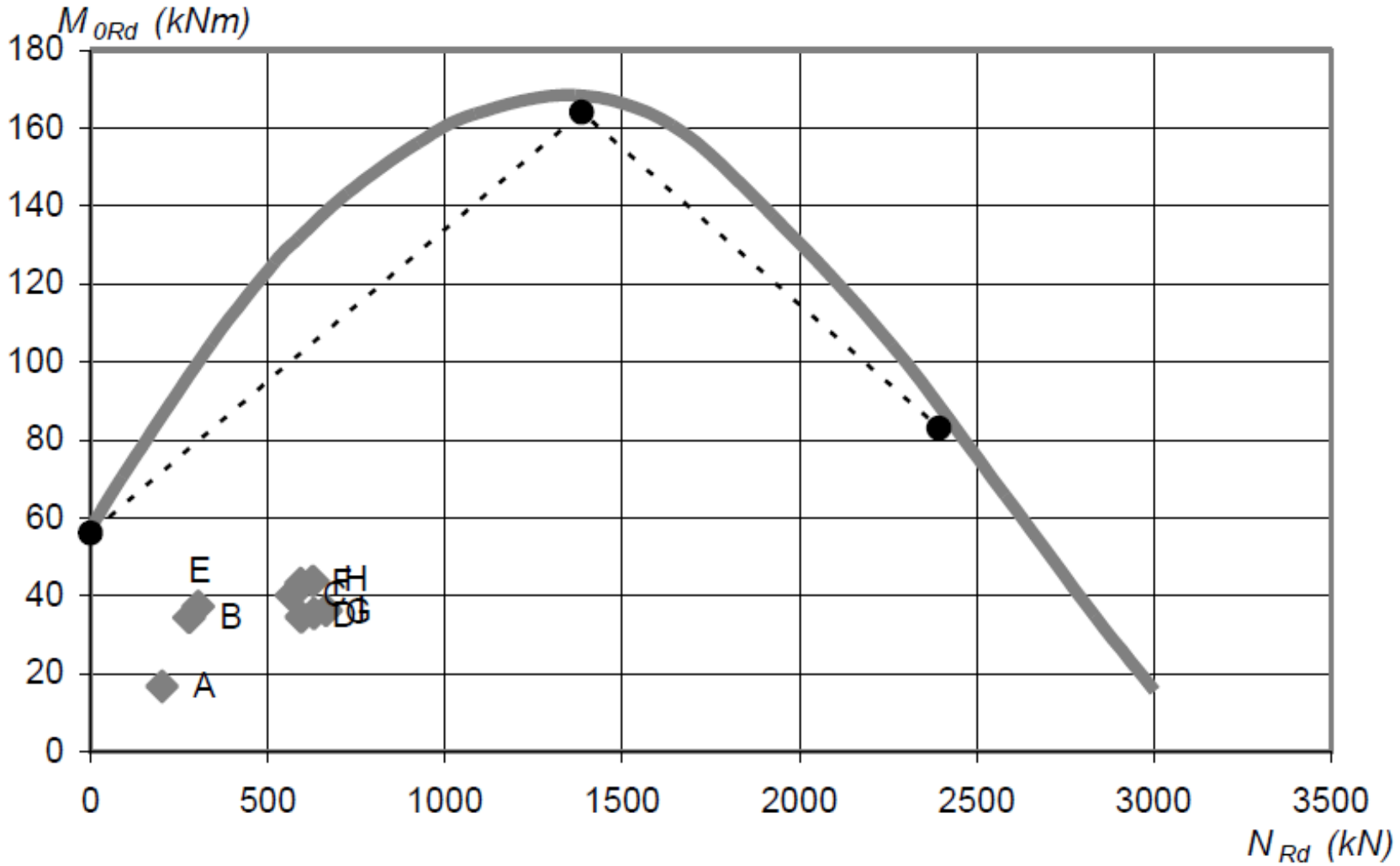
$$M_{0Ed} = N_1 e_1 + N_0 e_0 - N_2 e_2 + \frac{1}{8} w l^2$$

$$e_0 = 20mm$$
$$e_1 = 233mm$$
$$e_2 = 200mm$$

	N_1	N_0	N_2	w	N_{Ed}	M_{0Ed}
A	63	114	24	2,94	200	16,7
B	139	114	24	2,94	277	34,6
C	139	403	24	2,94	566	40,4
D	139	403	52	2,94	595	34,6
E	165	114	24	0,88	302	37,3
F	165	403	24	0,88	591	43,1
G	165	403	62	0,88	630	35,4
H	165	438	24	0,88	626	43,8
I	165	438	62	0,88	664	36,1

Figure 8-15: Summary of the load combinations of the column, strong axis

Combined normal force and moment – practical use of design diagrams for verification



What about SLS?

- Buckling analysis is an essential part of any ULS verification
- Deflection analysis of columns and walls in SLS is in principle the same as for beams
- Crack width calculations are typically not relevant for columns and walls as the normal force normally ensures an uncracked cross section